surplus resources may be required in the future. Cognisant approximations of the number of patients who will require hospitalisation, mechanical ventilation, ICU infrastructure, etc, in the very near future will significantly enhance the promptness of responses and mitigation strategies.

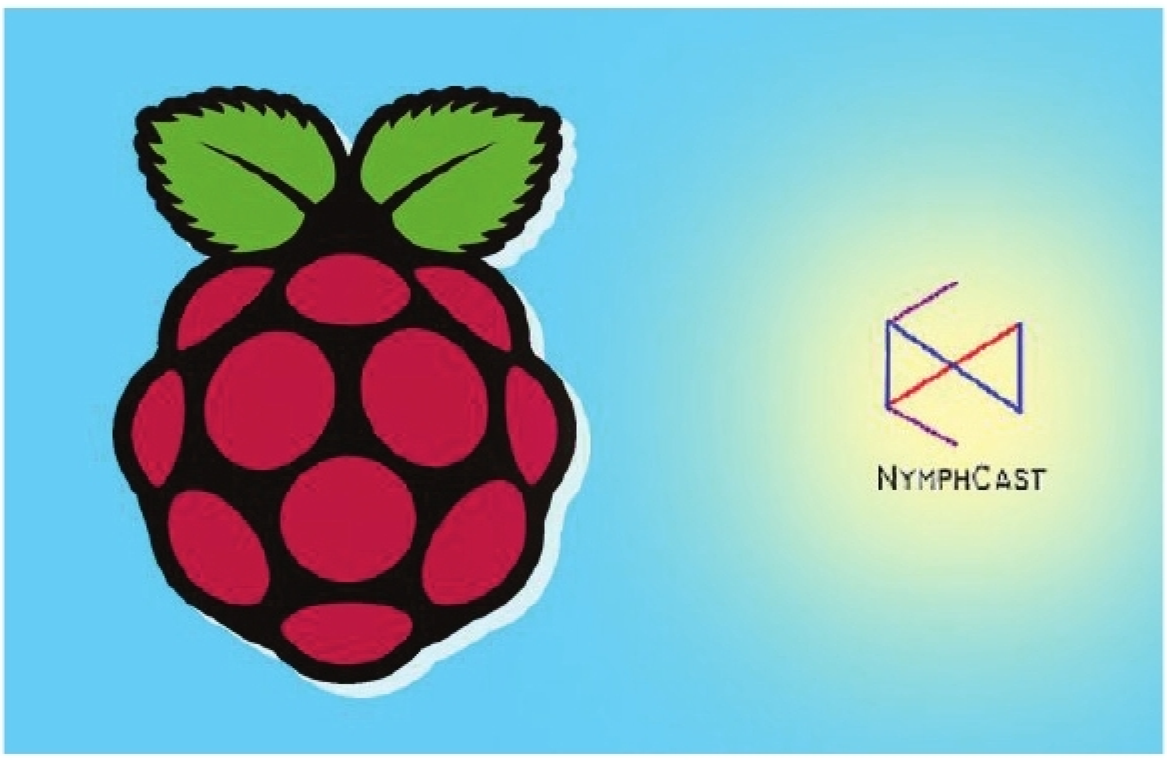
To this end, the Predictive Healthcare team at US based Penn Medicine has created a tool called CHIME that leverages SIR modelling to aid hospitals with capacity planning around COVID-19.

CHIME lets hospitals input information about their patients and adjust suppositions around the coverage and conduct of COVID-19. It runs a standard SIR model to predict the number of new admissions each day, accompanied by the day-to-day hospital survey. These predictions can be used to develop worst- and best-case scenarios to support capacity formulations.

The most significant feature in this model is the doubling time. It describes how quickly a disease spreads. Familiarities with other geographical contexts hint that the doubling time may range from three to 13 days, or more. The US, according to *Our World in Data*, has a four-day doubling rate. It does not paint a promising picture in the fight towards the corona virus. Dr Liz Specht guesstimated that even at a doubling rate of six, "all hospital beds in the US will be filled by about May 10."

Open source NymphCast lets Raspberry Pi to be used like Chromecast

Why buy a Chromecast stick when you can build your own? For those who do not know, Chromecast is a dongle for your television, connecting to the TV's HDMI port to add smart functions to your TV, like Netflix streaming. It's an HDMI dongle that lets one wirelessly 'cast' content to a device such as a TV.



Software engineer Maya Posch is transforming Raspberry Pis into streaming devices with her latest project NymphCast. This open source Chromecast alternative allows the user to stream video and audio to their television sets using a Raspberry Pi.

Lately, the project was shared on Reddit with news about its development over the last six months on GitHub. NymphCast is compatible with various Linux hardware with a ready-to-use image publicised just for the Raspberry Pi. The Raspberry device performs as a server for the NymphCast which can be regulated using a client, like as in a PC or a smartphone. Nymphcast is

Kubeflow 1.0 eases machine learning workflows with Kubernetes

Project co-authors Jeremy Lewi, Josh Bottum, Elvira Dzhuraeva, David Aronchick, Amy Unruh, Animesh Singh and Ellis Bigelow have made the following announcement on *Medium:* "Kubeflow's goal is to make it easy for machine learning engineers and data scientists to leverage cloud assets (public or on-premise) for (machine learning) workloads. With Kubeflow, there is no need for data scientists to learn new concepts or platforms to deploy their applications, or to deal with ingress, networking certificates, etc."

In 2017, Kubeflow was first introduced to the world at the annual Kubecon conference. Come 2020, its first major release is available. Kubeflow was developed to focus on two of the foremost concerns with machine learning projects: the requirement for integrated, end-to-end workflows, and the demand to make deployments of ML systems manageable and scalable. It allows data scientists to develop Ml workflows on Kubernetes and to manage them without worrying about how Kubernetes works.

Kubeflow is intended to manage each phase of any ML project—writing the code, assembling the containers, allotting the Kubernetes resources to run them, training the models, and providing predictions from those models.

According to Google, Kubeflow offers isolation, repeatability, resilience, and scale, not only for model training and forecast serving, but also for research and development purposes. Jupyter notebooks, a Kubeflow tool, can be process- and resource-limited, and can re-utilise configurations and data sources.

Several Kubeflow components are still under development and will be launched in the near future.